VERTICALLY INTEGRATED SYSTEMS

DOC.IV-7
EDITION: 2025-Q3

PRODUCT CATALOG

FIBER OTICS & PHOTONICS

2025

SCOPE: Optoelectronic Components & Solutions

APPLICATIONS: Optical Communication, Consumer & Automotive

KEY TECH: VCSELs, PIN Photodiodes, SiGe BICMOS

01. WELCOME TO VI SYSTEMS

VI SYSTEMS (VIS) DEVELOPS AND MANUFACTURES CUTTING-EDGE OPTOELECTRONIC COMPONENTS FOR COMMUNICATION, CONSUMER, AND AUTOMOTIVE APPLICATIONS. LOCATED IN THE HEART OF BERLIN, WE ARE A FABLESS COMPANY SPECIALIZING IN ULTRA-FAST SOLUTIONS FOR OPTICAL INTERCONNECTS AND SENSORS.

OUR PORTFOLIO INCLUDES VERTICAL CAVITY SURFACE-EMITTING LASERS (VCSELS) AND PIN PHOTODIODES DELIVERING SPEEDS UP TO 224 GBPS PER CHANNEL, ALONGSIDE DRIVER AND AMPLIFIER ICS OPERATING UP TO 56 GBPS. OUR LATEST GENERATION COMPONENTS ENABLE ENERGY-EFFICIENT 400 GBPS PAM-4 TRANSMISSION USING SHORT WAVE DIVISION MULTIPLEXING (SWDM).

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03. TECHNOLOGY & SERVICES

INTEGRATED OPTICAL SOLUTIONS

WE PAIR STATE-OF-THE-ART SIGE BICMOS INTEGRATED CIRCUITS WITH OUR ULTRAHIGH-SPEED VCSEL AND PIN COMPONENTS TO DELIVER OUTSTANDING PERFORMANCE.

WAFER MAPPING SERVICES

COMPREHENSIVE ELECTRICAL AND OPTICAL TESTING ON 2" TO 8" WAFERS AT TEMPERATURES UP TO 150°C.

HIGH-FREQUENCY TEST & CHARACTERIZATION

ELECTRO-OPTICAL PERFORMANCE MEASUREMENT WITH DATA MODULATION ANALYSIS UP TO 128 GBIT/S.

04. VCSEL TRANSMITTER MODULES

V25-850M

DATA RATE	28 GBPS (NRZ)
WAVELENGTH	850 NM
FIBER TYPE	50/125 MM

DATASHEET

V50-850M

DATA RATE	56 GBPS (PAM-4)
WAVELENGTH	850 NM
FIBER TYPE	50/125 MM

DATASHEET

VM100-850M

DATA RATE	112 GBPS (PAM-4)
WAVELENGTH	850 NM
FIBER TYPE	50/125 MM

DATASHEET

T56-850

DATA RATE	56 GBPS (NRZ)
WAVELENGTH	850 NM
FIBER TYPE	50/125 MM

DATASHEET

V25-1550M

WAVELENGTH	1550 NM
DATA RATE	25 GBPS (NRZ)
FIBER TYPE	9/125 MM

DATASHEET

05. OPTICAL RECEIVERS & PHOTODETECTORS

R56-850

INPUT WAVELENGTH	700–870 NM
DATA RATE (NRZ)	56 GBPS
FIBER TYPE	50/125 MM

DATASHEET

D30-850M

3 DB BANDWIDTH	> 30 GHZ
WAVELENGTH	840-1650 NM
FIBER TYPE	50/125 MM

DATASHEET

D60M FC

INPUT WAVELENGTH	400–1650 NM
3 DB BANDWIDTH	> 60 GHZ
FIBER TYPE	MMF/SMF

DATASHEET

06. INTEGRATED CIRCUITS (ICS)

HIGH SPEED VCSEL DRIVERS

A56-230C

DATA RATE	UP TO 100 GBPS
SUPPLY VOLTAGE	3.3 V
POWER DISSIPATION	200 MW

DATASHEET

A56-105C

DATA RATE	UP TO 56 GBPS
SUPPLY VOLTAGE	3.3 V
POWER DISSIPATION	105 MW

DATASHEET

T56-250C

DATA RATE	UP TO 56 GBPS (NRZ)
DIFFERENTIAL GAIN	3.0 KΩ
POWER DISSIPATION	150 MW

DATASHEET

08. ULTRA HIGH-SPEED VCSEL CHIPS

SERIES: VM100 (MM / MA-SM / QSM)

VM100 850

CHIP TYPE	MULTI-MODE
WAVELENGTH	840–860 NM
CONTACT	GSG

DATASHEET

VM100 880

CHIP TYPE	MULTI-MODE
WAVELENGTH	870–890 NM
CONTACT	GSG

DATASHEET

VM100 910

CHIP TYPE	MULTI-MODE
WAVELENGTH	900–920 NM
CONTACT	GSG

DATASHEET

VM100 940

CHIP TYPE	MULTI-MODE
WAVELENGTH	930–950 NM
CONTACT	GSG

DATASHEET

VM100 850 MA-SM

CHIP TYPE	MA-SM
WAVELENGTH	840–860 NM
CONTACT	GSG

DATASHEET

VM100 850 QSM

CHIP TYPE	QSM
WAVELENGTH	840–860 NM
CONTACT	GSG

DATASHEET

09. HIGH-SPEED VCSEL CHIPS

SERIES: V50 / VM50 / V25

V50 850

CHIP TYPE	MULTI-MODE
WAVELENGTH	840–860 NM
CONTACT	SG / GS

DATASHEET

VM50 940

CHIP TYPE	MULTI-MODE
WAVELENGTH	930–950 NM
CONTACT	GSG

DATASHEET

V25 940 HP MA

FEATURE	HIGH POWER (LIFI)
WAVELENGTH	930–950 NM
CONTACT	SG / GS

DATASHEET

VM50 850

CHIP TYPE	MULTI-MODE
WAVELENGTH	840–860 NM
CONTACT	GSG

DATASHEET

V25 850 HT

FEATURE	HIGH TEMP (125°C)
WAVELENGTH	840–860 NM
CONTACT	SG / GS

DATASHEET

10. HIGH-SPEED PHOTODIODES (PDS)

SWDM-OPTIMIZED

D40 SWDM

CONTACT TYPE	GSG
DIAMETER	20 MM
BANDWIDTH	~40 GHZ

DATASHEET

D35 SWDM

CONTACT TYPE	GSG
DIAMETER	25 MM
BANDWIDTH	~35 GHZ

DATASHEET

D30 SWDM

CONTACT TYPE	GSG
DIAMETER	~23 MM
BANDWIDTH	~30 GHZ

DATASHEET

D70 SWDM

CONTACT TYPE	GSG
DIAMETER	~25 MM
BANDWIDTH	~70 GHZ

DATASHEET

D400G

CONTACT TYPE	GSG
DIAMETER	~16 MM
BANDWIDTH	~60 GHZ (25Ω)

DATASHEET

11. CONTACT INFORMATION

FOR ADDITIONAL INFORMATION OR TO RECEIVE A QUOTATION, PLEASE CONTACT OUR SALES DEPARTMENT.

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